

## GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>Project</b>                 | FEED for Gas Gathering Pipelines & Associated Infrastructure |
| <b>Client</b>                  | Brass Fertilizer and Petrochemical Company Limited           |
| <b>Originating Company</b>     | Epic Atlantic Limited                                        |
| <b>Document Title</b>          | EPC Cost Estimate                                            |
| <b>Document Number</b>         | EA-BFPCL-GPMC-GGPL-FEED-PMC-RPT-003                          |
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| <b>Coordinator</b>             | Aloy Duru                                                    |
| <b>Security Classification</b> | Restricted                                                   |
| <b>Issue Date</b>              | 02 Feb 2023                                                  |

## EPC Cost Estimate

|       |            |                          |              |                 |                 |                  |
|-------|------------|--------------------------|--------------|-----------------|-----------------|------------------|
| A01   | 02.Feb.23  | Issued for Approval      | Onome Igbudu | Chijioke Anozie | Aloy Duru       |                  |
| R02   | 26.Aug.22  | Issued for Client Review | Onome Igbudu | Chijioke Anozie | Aloy Duru       |                  |
| R01   | 05.Jun.22  | Issued for Client Review | Onome Igbudu | Chijioke Anozie | Aloy Duru       |                  |
| R00   | 03.Jun.22  | IDC                      | Onome Igbudu | Chijioke Anozie | Aloy Duru       |                  |
| Rev # | Issue Date | Status Description       | Originator   | Reviewer        | Approver (Epic) | Approver (BFPCL) |

## Revision History

| Rev. No. | Date     | Description of Revision |
|----------|----------|-------------------------|
| R00      | 03.06.22 | For IDC                 |
| R01      | 05.06.22 | Issued for Review       |
| R02      | 26.08.22 | Re-issued for Review    |
| A01      | 02.02.23 | Issued for Approval     |

## Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)  
*for the subsequent project phases, post FEED, it is expected that:*
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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## SUMMARY PAGE

|                  |                                                     |                                                       |
|------------------|-----------------------------------------------------|-------------------------------------------------------|
| <b>Client:</b>   | Brass Fertilizer & Petrochemical Company Limited    |                                                       |
| <b>Project:</b>  | Gas Processing & Methanol Complex                   | <b>Project No.</b> 11998A                             |
| <b>Facility:</b> | Gas Gathering Pipelines & Associated Infrastructure | <b>Doc No.</b><br>EA-BFPCL-GPMC-GGPL-FEED-PMC-RPT-003 |

The administration of this document is in respect of EPC Cost Estimate of the FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

The document, in consideration of relational design elements & the FEED deliverables, forms the techno-commercial basis for the engineering, procurement, construction, and commissioning of Pipelines & Associated Infrastructure.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's Funding.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: EPC, LONG LEAD ITEMS, MATERIALS LIST, PIPELINE, MANIFOLD, PLATFORM

## REFERENCES

| Document No.                        | Description             |
|-------------------------------------|-------------------------|
| EA-BFPCL-GPMG-GGPL-FEED-GEN-RPT-001 | FEED Package Overview   |
| EA-BFPCL-GPMC-GGPL-FEED-MPPLST-005  | MTO for Long Lead Items |
|                                     |                         |
|                                     |                         |

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## Project Information, Terms, Abbreviations, and Definitions

### Project Information – General

|     |                                    |                                                                                                                                           |
|-----|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0 | Employer                           | BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED (BFPCL)                                                                                  |
| 2.0 | PMC                                | TATA CONSULTING ENGINEERS LIMITED (TCE)                                                                                                   |
| 3.0 | EPCC (CONTRACTOR)                  | CHINA TIANCHEN ENGINEERING CORPORATION (TCC)                                                                                              |
| 4.0 | Project Title                      | Gas Processing and Methanol Complex                                                                                                       |
| 5.0 | Location                           | Akabel, Brass Island.<br>The project site is about 85km South West of Port Harcourt, 70km South of Yenagoa and 90km West of Bonny Island. |
| 5.1 | State                              | Bayelsa State, Nigeria                                                                                                                    |
| 5.2 | Nearest Air Port and distance      | Port Harcourt Airport (PHC)<br>Bayelsa Airport (Under Construction)                                                                       |
| 5.3 | Nearest Port                       | Port Harcourt (Onne)                                                                                                                      |
| 5.4 | Nearest Railway Station & distance | -                                                                                                                                         |
| 5.5 | Highway and distance               | Connected through water way only. Road connection does not exist.                                                                         |
| 5.6 | Important Town and distance        | Yenagoa, Nigeria & 67 km                                                                                                                  |
| 5.7 | Altitude Above Mean Sea Level      | 9 m                                                                                                                                       |
| 5.8 | Atmospheric pressure               |                                                                                                                                           |
|     | Average                            | 1013 mbar                                                                                                                                 |
| 6.0 | Ambient Air Temperatures           | (Refer Note-5,6)                                                                                                                          |
| 6.1 | Maximum                            | 38°C                                                                                                                                      |
| 6.2 | Minimum                            | 18°C                                                                                                                                      |
| 6.3 | Average                            | 23°C                                                                                                                                      |
| 6.4 | Summer Design Wet Bulb Temp        | 28°C                                                                                                                                      |
| 6.5 | Winter Design Wet Bulb Temp        | 15°C                                                                                                                                      |
| 6.6 | MDMT                               | 15°C                                                                                                                                      |

## Project Information – General (Contd)

|      |                                                     |                                                                   |
|------|-----------------------------------------------------|-------------------------------------------------------------------|
| 6.7  | Swamp temperature                                   | 250C. (230C. to be used wherever a margin is required)            |
| 6.8  | Ground temperature                                  | 25 - 27.50C.                                                      |
| 6.9  | Black-bulb temperature:                             | 800C.                                                             |
| 7.0  | Relative Humidity                                   |                                                                   |
| 7.1  | Maximum                                             | 95 %                                                              |
| 7.2  | Minimum                                             | 51 %                                                              |
| 7.3  | Average                                             | 73 %                                                              |
| 8.0  | Rainfall Data                                       |                                                                   |
| 8.1  | Average annual rainfall                             | 3800mm                                                            |
| 8.2  | Max daily rainfall:                                 | 180mm                                                             |
| 8.3  | Mean max. Hourly rainfall                           | 100mm                                                             |
| 9.0  | Transport                                           |                                                                   |
| 9.1  | Name of the highway near which the plant is located | Connected through water way only. Road connection does not exist. |
| 9.2  | Access road                                         | -                                                                 |
| 9.3  | Railway (Gauge)                                     | -                                                                 |
| 10.0 | Wind Data                                           |                                                                   |
| 10.1 | Design Wind Velocity                                | 35.6 m/sec (Note-1)                                               |
| 10.2 | Wind Direction                                      | Note-2                                                            |
| 11.0 | Seismic Data                                        |                                                                   |
| 11.1 | Earthquake Zone                                     | Note-3                                                            |
| 12.0 | High Flood Level (HFL)                              | *                                                                 |
| 13.0 | Natural Ground Level                                | *                                                                 |
| 14.0 | Slope                                               | *                                                                 |
| 15.0 | Drainage                                            | *                                                                 |
| 16.0 | Design temperature for Electrical equipment         | 400C                                                              |

## Definition of Terms

| Terms                 | Definitions                                                                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BFPCL                 | Brass Fertilizer & Petrochemical Limited.                                                                                                                                                                                                                                        |
| SPDC                  | Shell Petroleum Development Company Nig. Limited                                                                                                                                                                                                                                 |
| Employer              | Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer. |
| Engineer              | The party that carries out the FEED Engineering of the project.                                                                                                                                                                                                                  |
| Manufacturer/Supplier | Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.                                                                                                                                                                  |
| Shall                 | The word 'shall' indicate a mandatory requirement.                                                                                                                                                                                                                               |
| Should                | The word 'should' indicate a recommendation.                                                                                                                                                                                                                                     |
| Measuring Range       | Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).                                                                                                                                |
| On-Line Instruments   | Instruments connected to process and utility lines or equipment via primary isolation valves.                                                                                                                                                                                    |
| Project               | Front End Engineering Design for Gas Gathering Pipelines (Phase1)                                                                                                                                                                                                                |

## 1. Introduction

### 1.1. Background Information

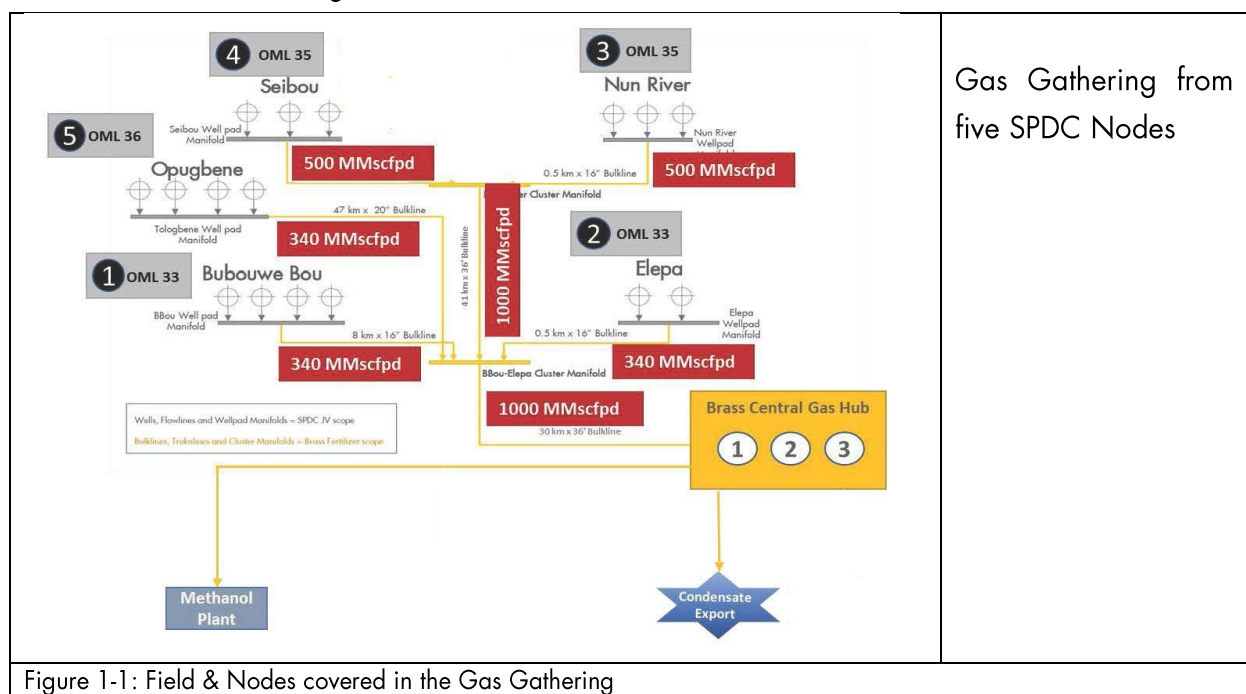
Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a greenfield methanol production facility in Akabel (Odioma), Brass LGA Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields, namely: Bubouwe Bou & Elepa, in OML 33, Nun River, Seibou, in OML 35, and Opugbene in OML 36.

The Project involves the development, construction, and operation of a 10,000 metric tonnes per day (MTPD) capacity methanol production plant comprising the following elements:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel, - as shown in Figures 1.1, 1.2, & 1.3.
- Product Export Lines and associated Offshore Single Buoy Mooring (SBM) for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) which are required for the operation of the processing plants.

Tata Consulting Engineers Limited (TCE) is the Project Management Consultant for the project.

#### The Nodal Gas Gathering Schematics





## The Nodal Gas Gathering Schematics (contd.)

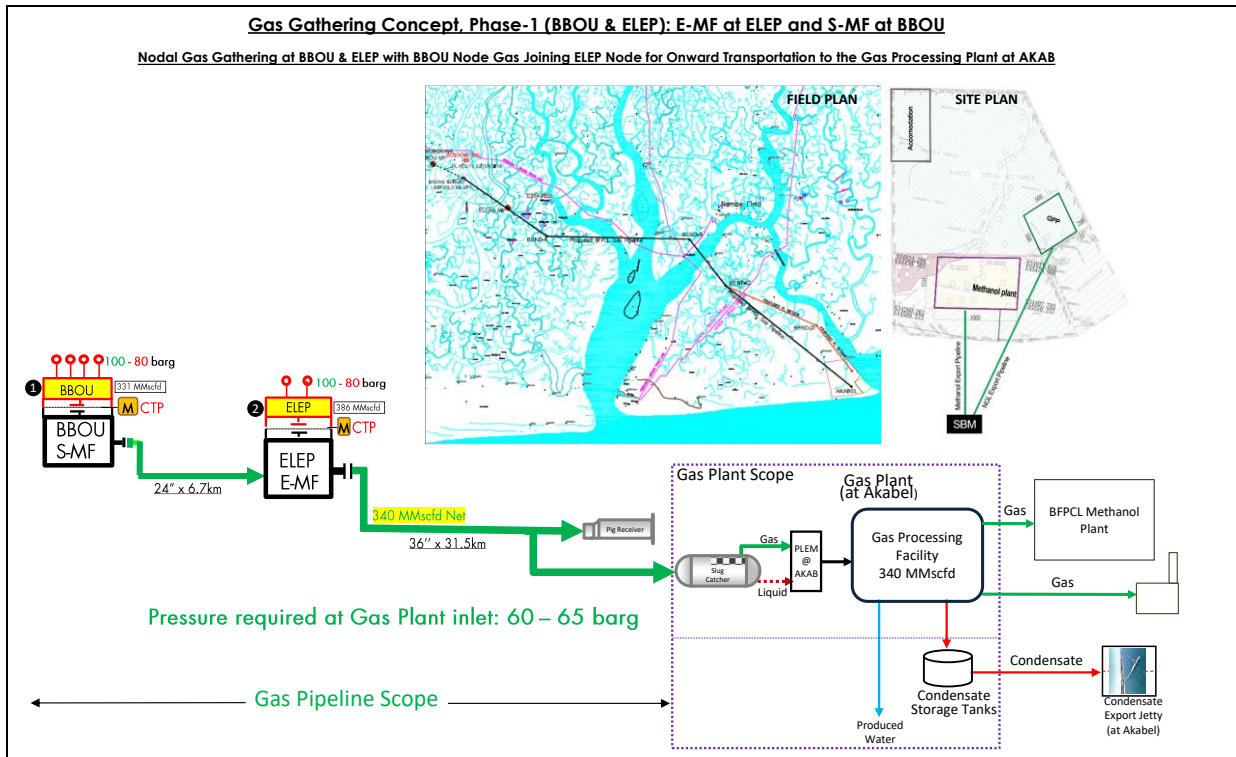
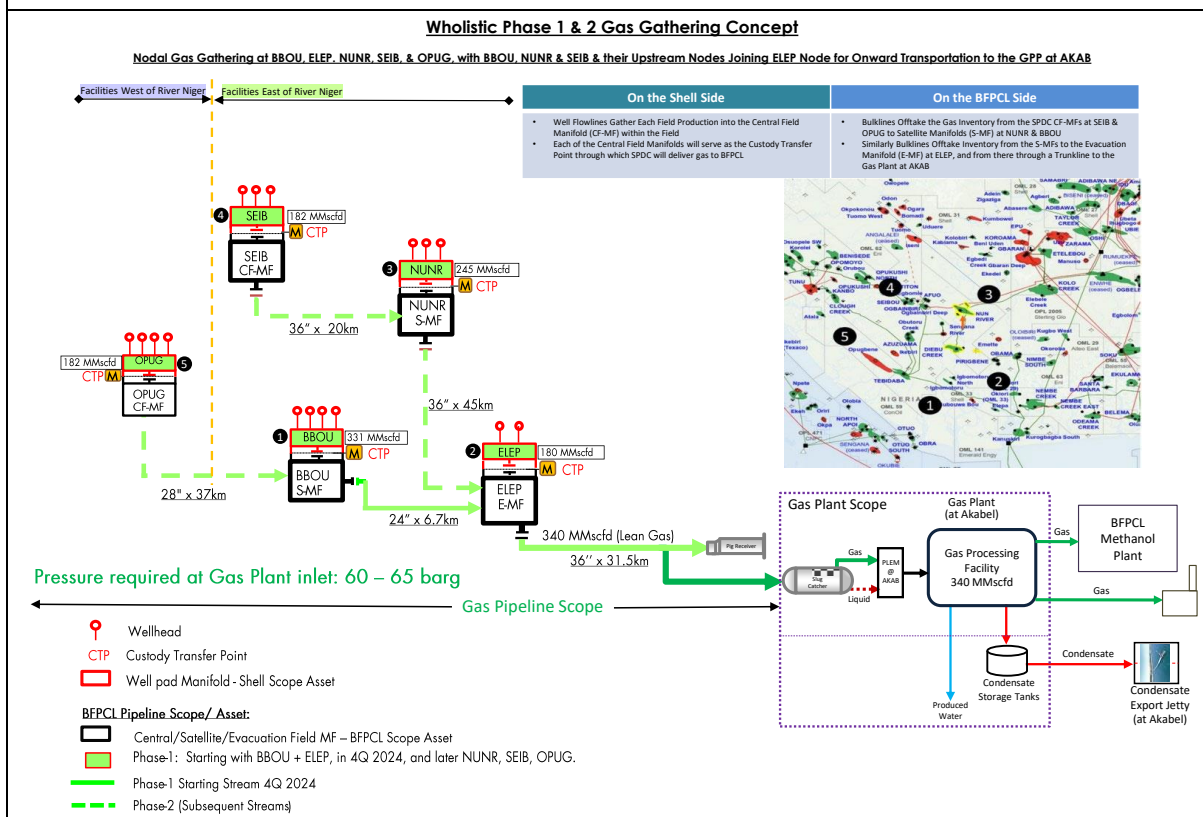


Figure 1-2: Gas Gathering Concept, Phase 1



Note: The 28" OPUG to BBOU is only a notional to ensure target pressure at AKAB, pending Phase-2 FEED

Figure 1-3: Wholistic Gas Gathering Concept, Phase 1 & 2

## 1.2. Purpose of Document

The purpose of this document is to provide a level 3 cost estimate to +10% accuracy for the purpose of progressing the project to funding and EPC tender, as well as present a benchmark for the evaluation of the commercial bid that will be received imminently.

The gas gathering will be in two phases, as shown in the schematics below:

- a. Phase 1 comprising gas gathering from Bubouwe Bou and Elepa fields, and
- b. Phase 2 comprising gathering from Nun River, Seibou, and Opugbene fields.

### 1.2.1. The Phase 1 Facilities

The phase 1 facilities (shown in Figure 1.1) are:

- BBOU – ELEP pipeline
- ELEP - AKAB Pipeline
- BBOU satellite Manifold
- ELEP evacuation Manifold, and
- The Pipeline End Manifold at AKAB.

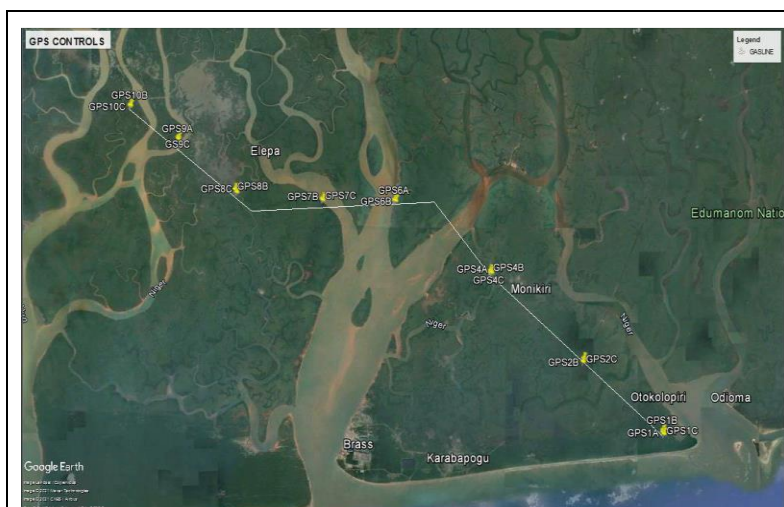
These phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

The pipeline route proposed for this phase 1 has been surveyed, and the survey report issued for the FEED. The phase 2 pipelines routes are yet to be defined and surveyed.

### 1.2.2. The Phase 2 Facilities

The phase 2 facilities comprise NUNR, SEIB, and OPUG Manifolds with their pipelines connecting to phase 1 facilities as shown in Figure 1.2.

## 1.3. The Phase 1 Pipelines Route / Right of Way



The pipeline route / Right of Way, running from BBOU through ELEP to AKAB is shown in the satellite image below

Figure 1-4: Satellite Image of the proposed BBOU – ELEP-AKAB Right of Way

## 1.4. Scope

The Facilities to be covered under project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

## 1.5. Design Battery limits

The battery limits for this Phase 1 FEED Project are:

### 1.5.1. BBOU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

### 1.5.2. ELEP Platform

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

### 1.5.3. Akabel Facility

- The Pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

## 2. Regulations, Basis and Assumptions

### 2.1. General

The development of the cost estimates has considered regulations, taxation, and Incoterms in developing activities & business processes that led to the cost estimates.

### 2.2. Key Regulations

| Regulation                                                                    | Function                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local Content Requirements                                                    | The provisions of the Nigerian Content Act apply to this project. The design includes strategies to comply with the Act and to maximize Nigerian local content throughout the life of the project. Exemptions / waivers have been sought where required. |
| Incoterms                                                                     | A set of internationally recognized rules which define the responsibilities of sellers and buyers, in respect of Cost, Insurance, Freight, and Taxes.                                                                                                    |
| Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended. | This is the principal statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.                                                                                                              |
| Mineral Oil (Safety) Regulations 1963, as amended.                            | These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.                                                                                                                          |
| Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)          | Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.                                                           |
| Oil and Gas Pipelines Regulations, 1995                                       | Details the regulations for the design, construction and maintenance of oil and gas pipelines.                                                                                                                                                           |
| Environmental Impact Assessment Act, CAP E12, LFN 2004.                       | Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment                                                                                                         |
| Federal Environment Protection Agency Act, CAP F13, LFN 2004.                 | Provides for the protection of the environment and sustainable development of Nigeria's natural resources.                                                                                                                                               |

### 2.3. Basis & Assumptions

The key basis & assumptions are the following:

- a. Detailed to arrive at  $\pm 10$  accuracy
- b. Base cost estimate containing no allowances for risk or inflation
- c. Reference to recent similar jobs
- d. Offers returned by Pipeline Construction Contractors
- e. Offers returned by Vendors

### 2.4. Cost Estimate Methodology

The methodologies followed are:

- a. Complete identification & definition of all constituent materials of the project and costs of the materials
- b. Complete identification & definition of all activities for the project's execution. These activities are either stand-alone or under a more major activity
- c. Complete with cost elements of logistics, Community Affairs (MOU, Land Acquisition, Community Engagements, etc) to the extent foreseeable, Personnel Cost, Permits and Survey, HSE (Induction Medical)
- d. Peer review
- e. Provision of 10% Contingency or Shortfall of Bulk Materials to cater for errors & omissions

### 3. Cost Summaries

#### 3.1. The Cost Estimate

The cost estimate for the entire Pipeline System using the conventional open cut method is **USD279,990,430** with breakdown and details as shown below. Site services & Camp Management cost is added as it wasn't accounted for in the previous Cost Estimate report of 25<sup>th</sup> August 2022.

| s/n | Item Description                                           | BOQ or Specification                   | Cost        |             | 432 NGN / USD |             |
|-----|------------------------------------------------------------|----------------------------------------|-------------|-------------|---------------|-------------|
|     |                                                            |                                        | NGN         | USD         | F \$          |             |
| I   | DIRECT COST                                                |                                        |             |             |               | 240,986,350 |
| 1   | Pipeline System                                            |                                        |             |             | 238,781,049   |             |
| 1.1 | Line Pipes                                                 |                                        |             | 44,024,644  |               |             |
| 1.2 | Key Equipment                                              |                                        |             | 1,249,850   |               |             |
| 1.3 | Other Equipment                                            |                                        |             | 48,250      |               |             |
| 1.4 | Bulk Materials (Mech, Civil, Electrical, Instr & Telecoms) |                                        |             | 9,760,717   |               |             |
| 1.5 | Structural Steel                                           |                                        |             | 1,882,649   |               |             |
| 1.6 | Cable & Electric                                           |                                        |             | 4,490,086   |               |             |
| 1.7 | Civil Work                                                 |                                        |             | 11,219,858  |               |             |
| 1.8 | MEI Installation Work                                      |                                        |             | 166,104,996 |               |             |
| 2   | Consumable for Initial Filling                             |                                        |             | 500,000     | 500,000       |             |
| 3   | Incidental Cost of Eqpt & Materials                        |                                        |             |             | 1,705,300     |             |
| 3.1 | Ocean Freight & Ocean Shipping Insurance                   |                                        |             | 1,076,828   |               |             |
| 3.2 | Local Transportation to Site                               |                                        | 271,500,000 |             |               |             |
| II  | INDIRECT COST                                              |                                        |             |             |               | 33,500,000  |
| 4   | Pre-Engineering Survey                                     | Pre-Engineering Survey Works heet      |             |             | 1,000,000     |             |
| 5   | Detail Engineering Design                                  | Detail Engineering Design Works heet   |             |             | 2,500,000     |             |
| 6   | Project Management                                         | Project Management Works heet          |             |             | 5,000,000     |             |
| 7   | Procurement Service                                        | Procurement Services Works heet        |             |             | 1,500,000     |             |
| 8   | Construction Management                                    | Construction Management Works heet     |             |             | 6,000,000     |             |
| 9   | Insurance Fee                                              | Insurance Fee Works heet               |             |             | 2,000,000     |             |
| 10  | Commissioning                                              | Commis sioning Works heet              |             |             | 4,000,000     |             |
| 11  | Permits Fee                                                | Permits Fee Works heet                 |             |             | 500,000       |             |
| 12  | HSE Fee including community                                | HSE Fee including community Works heet |             |             | 9,000,000     |             |
| 13  | Training                                                   | Training Works heet                    |             |             | 2,000,000     |             |
| 14  | Site Services & Camp Management                            |                                        |             |             | 5,504,080     | 5,504,080   |
|     | Total                                                      |                                        | 271,500,000 | 240,357,877 | 279,990,430   | 279,990,430 |

### 3.2. Estimate for Continuous HDD

The estimate for continuous HDD is **USD 399,831,430** derived from the open cut lay by adding the HDD thrust boring cost, and discounting for HDD at river crossing already covered in the open cut method, and the following works that are not required by the HDD method.

1. Trenching & Backfilling
2. Concrete Weight Coating
3. Differential Cost of Welding heavier concrete coated pipe,

The cost estimate is summarised in the steps and building blocks below.

summary of the projected cost is shown below.

Table 3.2 Cost Estimate for the Continuous HDD constructed from the Conventional Open Cut Method.

| s/h   |                                         | Rate  | Qty    | Unit | USD         |
|-------|-----------------------------------------|-------|--------|------|-------------|
| 1     | Base Cost (open Lay)                    |       |        |      | 279,990,430 |
| 2     | Add Cost of HDD                         | 4,000 | 38,200 | m    | 152,800,000 |
|       | Total                                   |       |        |      | 432,790,430 |
| 3     | Works to the Credit of Continuous HDD   |       |        |      |             |
|       | HDD at Crossings on the Open Cut Method |       |        |      | 16,830,000  |
|       | Trenching & Backfilling                 |       |        |      | 3,820,000   |
|       | Concrete Weight Coating                 |       |        |      | 2,350,000   |
|       | Differential Cost of Welding            |       | 10%    |      | 9,959,000   |
|       | Total                                   |       |        |      | 32,959,000  |
| hence | Net Cost of HDD                         |       |        |      | 399,831,430 |

## 4. Attachment – Cost Estimate Workbook

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The cost estimate workbook is attached and embedded below.

|   |                               |                                                                                                                            |
|---|-------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1 | Cost Estimate Master Workbook | <br>Cost Estimate<br>Master Work Book_1 |
|---|-------------------------------|----------------------------------------------------------------------------------------------------------------------------|